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MILITARY SPECIFICATION

LIFE PRESERVER, INFLATABLE, CEMENTED AIRCREWMAN (WITH ARAMID CASING), TYPE LPU-23B/P

This specification is approved for use within Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for one type of aircrewman inflatable life preserver without automatic inflation device designated as Type LPU-23B/P.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, and standards . The following specifications, and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

UU-P-268	–	Paper, Kraft, Wrapping
UU-T-81	–	Tag, Shipping and Stock
PPP-B-636	–	Box, Shipping, Fiberboard
PPP-T-76	–	Tape, Packaging, Paper, (For Carton Sealing)

Military

MIL-P-116	–	Preservation-packaging, Methods of
MIL-C-19002	–	Cloth, Coated, And Tape, Coated Cloth-polychloroprene On Nylon, Pneumatic Life Preserver
MIL-I-23145	–	Inflation Assemblies, Life Preserver

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in Improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department, (SESD),(Code 93), Lakehurst, NJ 08733-5100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS (Continued)

Military (Continued)

MIL-C-25369 - Cylinders, Carbon Dioxide, For Inflation of Pneumatic Life Preservers.

MIL-O-81375 - Oral Inflation Assemblies, Survival Equipment, Inflatable

MIL-C-81814 - Cloth, Twill, Aramid, High Temperature Resistant

MIL-H-85478 - Handle, Beaded, Inflation Assembly

STANDARDS

Federal

FED-STD-751 - Stitches, Seams, and Stitchings

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Markings of US Military Property

2.1.2 Government drawings and publications. The following Government drawings and publication form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Naval Air Systems Command

68A73D21 - Life Preserver Assembly Index, Inflatable Aircrewmen, Type LPU-23B/P

68A73H1 Life Preserver Assembly, Aircrewmen, Inflatable, Aramid, Cemented, Type LPU-21C/P and Type LPU-23B/P

68A73D2 Pouch Assembly, Dye Marker

68A73D3 Pouch Assembly, flare

68A73H4 Flotation Assembly, Type LPU-21C/P and Type LPU-23B/P

68A73H6 Bladder Assembly, Flotation

68A73D7 Attaching Item, Hardware

68A73D8 Patches, Assemblies, Flotation

68A73C9 Patches, Bladder

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DRAWINGS (Continued)

Naval Air Systems Command (Continued)

68A73C11	Identification Patch
68A73H12	Patterns, Bladder
68A73D19	Baffle Assembly
62A119D1	Oral Inflation Assemblies, Inflatable Survival Equipment
62A120H1	Manifold Stem Assembly and Details, For Life Preservers
67A319C15	Patch, Inspection Record
67A319D18	Check Valve Assembly
67A319D19	Hardware, Harness, Details
975AS103	Casing Assembly, Aramid, Type LPU-21C/P and Type LPU-23B/P
975AS105	Main Panel Sub-Assembly, Casing
975AS106	Back Panel Sub-Assembly, Casing
975AS11O	Casing Details, Miscellaneous
975AS112	Casing Detail, Hardware
975AS118	Pocket, Protective
975AS119	Cover, Protective
975AS120	Cover, Locking Pin
975AS121	Handle, Beaded, Inflation Assembly
975AS200	Casing Patterns, Type LPU-21C/P and Type LPU-23B/P

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND

NAVAIR 13-1-6.1 Aviation Crew Systems Manual Inflatable Survival Assembly of

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. The Type LPU-23B/P aircrewmen inflatable life preservers furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3, 4.3.1, 6.3 and 6.3.1).

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Materials and components. The materials and components shall conform to applicable specifications, standards, and drawings as listed or required herein. Unless otherwise specified in the contract, the manufacturer shall certify that the materials and components, except for the metallic parts used in the construction of the life preservers, shall have been manufactured not more than 18 months prior to the date of delivery of the life preservers (see 6.2.1j and 6.2.2).

3.4 Design and construction. The design and construction of the LPU-23B/P life preserver shall be in accordance with drawing 68A73D21.

3.4.1 Construction details. All details for the component parts and sub-assemblies, not specified herein, shall be in accordance with the applicable detail drawings.

3.4.1.1 Patterns. The casing for the life preserver shall be cut in strict accordance with the drawing 975AS200 furnished by the Government (see 6.2.3). The Government pattern shall not be altered in any manner and shall be used for making the working patterns. The working patterns shall be identical to the Government furnished pattern drawing.

3.4.2 Inflatable compartments. The Type LPU-23B/P life preserver shall consist of two independent inflatable compartments. Each compartment shall be capable of being inflated mechanically with carbon dioxide and orally through an oral inflation valve and tube. The deflated flotation compartment shall be attached to and contained in a casing, which shall be worn around the waist, up the back, and around the neck. The life preserver in its casing shall conform in appearance to drawing 68A73H1.

3.4.3 Cut edges. All the cut edges of the uncoated nylon tapes (including the fastener tapes) and webbings shall be seared, prior to the fabrication of the life preserver, to prevent fraying. No sharp edges shall be formed.

3.4.4 Use of the adhesive. The color of the adhesive shall be natural (unpigmented). In all the cementing operations, the surface to which the adhesive is to be applied shall be thoroughly cleaned with a suitable solvent in such a manner that the dusting materials (zinc stearate or talc) or any other surface contaminants are removed. Care should be exercised to ensure that the coating and the base undercloth are not damaged and the adhesion between both is not impaired in any way by prolonged exposure to the solvent. The solvent used shall evaporate completely prior to the application of the adhesive and shall leave no residue. Cemented areas shall not contain trapped air, channels, or excessive wrinkles. The adhesive shall be controlled to ensure that old adhesive or adhesive which has partly or completely polymerized is not used. A fresh batch of the adhesive shall be used at least every eight hours during the course of manufacture. Containers for the adhesive shall be free from congealed adhesive before being refilled.

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3.4.5 Cementing of the seams, seam tapes, patches, and attachments. The construction of the seams and sealing areas and the attaching of the seam tapes, patches, and attachments shall be accomplished by utilizing the technique and precautions outlined in 3.4.4 so that, prior to submission of the assembled lIfe preserver, the adhesive shall have developed its optimum bonding properties and the adherence of all such seams, sealed areas, seam tapes, patches, and attachments shall comply with the requirements of this specification. All the seams, seam tapes, patches, sealed areas and attachments shall be secured by the adhesive as specified in the applicable drawings. The seams shall be covered on the inner and outer sides of the life preserver, as applicable, with the seam tape, as specified in the applicable drawings. All the seam tapes shall be applied to the seams without tension and shall be applied so that an equal amount is on each side of the seam edge which it covers. The seams shall be formed without tension and the patches and attachments shall be applied to the life preserver without tension. The adhesive shall be applied in a straight line parallel to the edges of the seam tapes, patches, and attachments and shall extend from just being visible to not greater than 3/8 inch beyond the edges of the seam tape, patch and attachment. The seam tapes which seal circumferential seams, or where one end of the seam tape comes in contact with the other end of the same seam tape or another seam tape, shall overlap on itself, at the ends, 3/4. plus 1/2, minus O, inch. The adhesive, when dry or cured, shall Present a neat and uniform appearance. The adhesive shall not be allowed to remain in clots nor permitted to extend in such a manner as to result in localized stiffness which might result in discomfort to the wearer, when the life preserver is deflated or inflated. The adhesive, upon drying or curing, shall not cause the cloth, seam tapes, patches, and attachments to shrink or pucker at any point on the life preserver. The adhesive shall not be allowed to run over the edges of the life preserver to form stiff or hard needle points and all the edges around the entire periphery of the life preserver, both inside and out, shall be free from sharp points or ragged edges. The life preserver shall be free from congealed masses of the adhesive and spots or stains resulting from excess adhesive.

3.4.6 Seams and stitching. No stitching shall pass through any air retaining compartment. Sewing shall only be used in the construction of the accessory parts but not in their final attachment to the life preserver. All the machine stitching in the accessory parts shall be accomplished with Stitch Type 301 FED-STD-751, eight to ten stitches per inch. Each row of stitching shall be straight and parallel to the seam edge. The straightness of the stitching in any row shall be maintained within a tolerance of plus or minus l/32 inch. The ends of the stitching shall be backstitched, by overlapping on itself, not less than 1/2 inch. Thread breaks, skips, and run-offs shall be overstitched not less than one inch. The thread tension shall be maintianed so that there shall not be any loose or tight stitching and the lock shall be embedded in the materials sewn together. no seam shall be twisted, puckered, or pleated and no portion of the accessory parts shall be puckered, or pleated and no portion of the accessory parts shall be caught in an unrelated operation and seam. All the thread ends shall be trimmed to a length of 1/4 or 3/8 inch. The seam edges shall be properly forced out and shall not contain any folds. No cut end of the nylon tape binding the edges of the casing shall be visible.

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3.4.6.1 Bartacking. The number of stitches per bartack shall be based proportionally on 1.4 inch long bartack containing not less than 14 stitches. The length and loation of each bartack shall be as specified in the applicable drawing.

3.5 Attachment and connections.

3.5.1 Manifold stem and check valve. Each compartment of the flotation assembly shall contain a mainfold stem assembly, conforming to drawing 62A120H1 and a check valve assembly conforming to drawing 67A319D18.

3.5.2 Oral Inflation assembly. The oral inflation assembly conforming to MIL-O-81375 shall be installed in each waist lobe of th life preserver flotation assembly as shown in drawing 68A73H6. The type, class and style shall be as specified on drawings 68A73H6 and 62Al19D1. Bladder and baffles shall be as specified in drawings 68A73H12 and 68A73D19.

3.5.3 Belt portion. The belt portion of the life preserver shall be provided with "D" rings for attaching the pouches. Harness hardware details shall be in accordance with drawing 67A73D3.

3.5.4 Dye marker retaining pouch. The pouch assembly for retaining the dye marker shall be fabricated as specified in 68A73D2.

3.5.5 Flare retaining pouch . The pouch assembly for retaining the flare shall be fabricated as specified in 68A73D3.

3.5.6 Warning tag. A red warning tag conforming to UU-T-81, Type B, not smaller than Size 4, and strung with twine, shall be securely tied to the left belt-connecting assembly of the LPU-23B/P life preserver. The tag shall contain the following information.

WARNING

This life preserver shall not be used until subjected to the calendar inspection requirements of NAVAIR 13-1-6.1, Aviation-Crew Systems Manual.

3.5.7 Patches attaching Items and protective covers. Patches, attaching Items and protective covers shall be in accordance with drawings 68A73D8, 975AS118 and 975AS119, respectively.

3.5.8 Beaded Handle Inflation Assembly. Beaded handle inflation assemblies conforimg to MIL-H-85478 and Dug 975AS121 shall be provided for each compartment of the inflation assembly.

3.6 Dimensions. The dimensions of the life preserver, components, and the casing and components shall be as specified in the specified applicable drawings, Including 975AS103, 975AS11O, 975AS112 and 975AS200.

3.7 Color. The color of the life preserver, casing, and components shall be as specified in the applicable drawings.

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3.8 Markings. Markings shall be in accordance with MIL-STD-130. The ink, color, size, and locations shall be as specified on applicable drawings. Letters and numerals 1/4 inch or less in height shall not be stenciled. The markings shall be thoroughly dry prior to packing. Serial numbers shall be as specified in 3.8.2.

3.8.1 Identification. The size of identification letters shall be in accordance with the applicable drawings. An identification patch shall be located on the flotation bladder assembly as shown on NAVAIR drawing 68A73H4. The construction and markings of the flotation bladder assembly Identification patch shall be in accordance with drawing 68A73C11. Identification markings for the main panel casing assembly shall conform to the requirements of drawing 975AS105. The date of manufacture and serial numbers, as applicable, may be rubber stamped (see 3.8.2).

3.8.2 Serial numbers. The results of the pressure inspection (4.7.3) and the leakage Inspection (4.7.4), shall be identifiable by an assigned serial number which shall be marked on the life preserver as specified in 3.8. Serial numbers shall be individually assigned by the manufacturer and shall consist of a block of consecutive numbers to cover the entire acquisition quantity.

3.8.3 Inspection record patch. The inspection record patch shall be located on the collar lobe of the flotation assembly. The construction, location and markings shall be as specified in drawings 67A319C15 and 68A73H4.

3.8.4 Warning Information. Naming information shall be marked on the life preserver as specified in 3.8 and drawings 975AS105, 975AS106, and 975AS200.

3.9 Performance requirements.

3.9.1 Operation (carbon dioxide). When the life preserver is inflated for operations test as specified in 4.7.2, design shape shall be attained within thirty seconds. There shall be no leakage of carbon dioxide, no hindrance to the flow of the carbon dioxide, or evidence of material or constructional failure in any respect. The seams of attachments shall not pucker when the compartments are inflated or deflated. Upon completion of this test and examination, and while still Inflated with carbon dioxide, the preserver shall be subjected to the buoyancy test (3.9.4 and 4.7.5), if required. If the buoyancy test is not required, the oral inflation assemblies will be inspected for conformance to the requirements of 3.9.4.

3.9.2 Pressure. When inspected as specified in 4.7.3, the pressure in each compartment shall be not less than 4.5 pounds per square inch gage (psig). All the seams, sealed areas, and cemented attachments shall remain perfectly intact and shall show no Indication of separation. There shall be no evidence of constructional or material failure in any respect. The seams or attachments shall not pucker when the compartments are inflated or deflated.

3.9.3 Leakage. When Inspected as specified in 4.7.4, the pressure In each compartment shall be not less than 1.60 pounds per square Inch gage psig. All the seams, sealed areas, and cemented attachments shall remain perfectly Intact and shall show no Indication of separation. There shall be no evidence of constructional or material failure In any respect. The seams or attachments shall not pucker, when the compartments are Inflated or deflated.

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3.9.4 Bouyancy. When Inspected as specified in 4.7.5, the life preserver shall support a 60 pound steel weight without the entire life preserver sinking below the surface of the water. Upon completion of the buoyancy inspection, each compartment shall be completely deflated through the respective oral inflation assemblies. During the deflation the oral Inflation assembly shall be checked not less than three times for operation without difficulty, restriction of gas flow, positive shut-off and integrity.

3.10 Workmanship. After completion of the final assembly, the life preservers shall be thoroughly cleaned and all loose thread, lint, and foreign matter shall be removed. The grommets and snap fasteners shall be clinched without distortion, damage, splitting, or cutting of the cloth. The metal components shall not be misaligned nor contain any sharp edge, crack, nick, burr, or sliver. The life preserver shall not contain any non-specified hole, abraded area, tear, cut, mend, or needle chew. The life preservers shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability, or durability. The life preservers shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or puchase order, the contractor is responsible for the performance of all inspection requriements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requriements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall Inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does It commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. Qualification inspection (see 4.3)

2. First article inspection (see 4.4)

3. Quality conformance Inspection (see 4.5)

4. Quality Conformance Verification Inspection (see 4.5.1.3).

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4.3 Qualification.

4.3.1 Samples. The qualification inspection samples shall consist of two life preservers. The qualification samples shall be forwarded as specified in 6.3. The samples shall be plainly identified by securely attached durable tags marked with the following information.

a. Samples for qualification Inspection
b. LIFE PRESERVER, INFLATABLE, AIRCREWMEN, TYPE LPU-23B/P
c. Manufacturer's designation or number
d. Name of manufacturer
e. Submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-L-85347 (applicable revision letter, including any amendment number) (AS) under authorization (reference authorizing letter and number) (see 6.3).

4.3.2 Inspection The qualification inspection of the LPU-23B/P life preserver shall cons st of all the examinations and tests specified in table r. The examinations and tests shall be performed in the sequence listed.

4.3.3 Rentention The retention of qualification shall consist of periodic verification to determine compliance of the qualified LPU-23B/P life preserver with the requirements of this specificaton. Periodic verification shall be by certification by the manufacturer and shall be at intervals of not more than two years.

4.4 First article.

4.4.1 First article samples. Unless otherwise specified in the contract, as soon as practicable after award of the contract or purchase order, the manufacturer shall submit three life preservers. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these life preservers from contract, submission of further first article samples on the new contract may be waived at the discretion of the contracting activity (see 6.2. le). Approval of the first article samples or the waiving of the first article inspection does not exempt the requirements for performing the quality conformance inspection. The first article samples shall be forwarded as specified in the contract or purchase order (see 6.2.1f).

4.4.2 First article sample approval. Upon completion of the first article inspection program, recommendations and comments pertinent for use In monitoring production, will be forwarded by the Government Activity responsible for the inspection program (see 6.2.1f) to the Contracting Activity. One approved life preserver will be returned to the manufacturer for use in monitoring production. One life preserver will be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract. The other life preserver will be retained by the inspection laboratory for reference and will be returned to the manufacturer with the samples from the final lot as required by 4.5.1.3.

4.4.3 First article Inspection. The first article Inspection of the Type LPU-23B/P life preserver shall consist of the examinations and tests specified in table II. The examinations and tests shall be performed In the sequence listed.

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4.5 Quality conformance inspection. The sampling and inspection levels shall conform to MIL-STD-105. The quality conformance inspection shall consist of the following:

a. Visual examinations of the life preservers
b. Operation (carbon dioxide)
c. Pressure
d. Leakage
e. Bouyancy
f. Dimensional check of the life preservers
g. Preparation for delivery

4.5.1 Sampling.

4.5.1.1 Inspection lot.

4.5.1 .1.1 Adhesive. An inspection lot size shall be expressed in units of one batch of the adhesive. An Inspection lot shall consist of all the batches of the adhesive used by the life preserver manufacturer during one day's production of the life preservers (see 3.4.4). The sample unit shall be one batch of the adhesive used during one day's production of the life preservers.

4.5.? .1.2 Life preservers. An inspection lot size shall be expressed in units of one life preserver made essentially under the same conditions and from the same materials and components. The sample units shall be one life preserver.

4.5.1 .1.3 Preparation for delivery. An inspection lot size shall be expressed in units of one fully prepared shipping container, including life preservers, fully prepared for delivery from essentially the same materials and components. The sample units shall be one shipping container, including life preservers fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examination of the components and the life preservers and preparation for delivery. The sample size, acceptance criteria, tests, and examinations required for the components, the life preservers of the preparation for delivery shall be as specified in table III and 4.5.1.3.

4.5.1.3 Quality conformance verification inspection. Upon completion of the tests and examinations specified in 4.5.1.2, a random sample shall be selected from each lot in accordance with MIL-STD-105. Inspection Level S-3, (see 6.2.lk). The sample size shall be based only on the applicable sample size code letter corresponding to the Inspection Level S-3. Each life preserver selected as a sample unit shall be identified by its assigned serial number (see 3.8.2), and shall be forwarded to the Government Laboratory specified In the procurement document (see 6.2.lg), for the following tests and examinations (listed sequence mandatory):

TESTS AND EXAMINATIONS

OPERATION	(4.7.2)
PRESSURE	(4.7.3)
LEAKAGE	(4.7.4)
VISUAL EXAMINATION	(4.7.1.1)
DIMENSIONAL CHECK	(4.7.1.1)

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The serial numbers of the units in the lot, represented by the sample units, shall be furnished to the Government laboratory. The initial cost of tests and examinations will be borne by the Government, however, the cost of tests and examinations of sample units resubmitted from a reworked lot, or from a new lot which may be necessitated by the rejection of a previous lot, shall be borne by the contractor. Sample units from a rejected lot shall not be resubmitted for tests and examinations without the approval of the contracting officer. Upon completion of testing, the sample units will be returned to the contractor, at the contractor's expense. The Government activity responsible for conducting the inspection program (see 6.2.lg) shall report the results of tests and examinations to the designated inspection and acceptance office specified in the procurement document. Final acceptance of the lot from which the sample units were selected shall be based upon successful completion of the inspection program by the cognizant Government Quality Assurance Representative/Specialist; applying the applicable acceptance criterias specified in table 111.

4.6 Inspection conditions.

4.6.1 Atmospheric conditions. Unless otherwise specified in the contract, all the inspections required by this specification shall be coducted at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77 ± 18 degrees Fahrenheit (25 ± 10 degrees Celsius). If the final values of the ambient temperature or barometric pressure at the end of the four-hour air pressure Inspections are different from the initial values recorded at the start of the inspection, the following corrections shall be made to the final pressure readings in psig.

4.6.1.1 Temperature correction. For each degree Fahrenheit rise in temperature 0.03 psig shall be added to the final pressure reading. For each degree Fahrenheit drop in temperature, 0.031 psig shall be added to the final pressure reading. The corresponding correction per degree Celsius is 0.056 psig.

4.6.1.2 Barometric pressure correction. For each 0.1 inch of mercury rise in barometric pressure, 0.049 psig shall be added to the final temperature-corrected pressure reading. For each 0.1 inch of mercury drop in barometric pressure, 0.049 psig shall be subtracted from the final temperature-corrected pressure reading.

4.6.2 Pressure measurement. The pressure shall be measured by means of a mercury manometer or low pressure gauge calibrated in tenths psig or fifths inches of mercury. Inches of mercury can be converted to psig by multiplying the inches of mercury by 0.49.

4.6.3 Inspection area and equipment. The area, in which the preservers are Inspected, shall be adequately protected to preclude damage to the units. The area and Inspection equipment shall be free of sharp or rough edges, burrs, protrusions, etc., which will cut, tear or damage the life preserver or the components.

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4.6.4 Air. When use of air is specified in an inspection, the air shall not contain any oil or condensed water vapor.

4.7 Inspection methods.

4.7.1 Visual examination.

4.7.1.1 Life preservers. Every life preserver assembly shall be visually examined and dimensionally checked for defects to determine conformance to this specification. The Classification of Defects as specified in (table IV) and the List of Defects as specified in (table V) shall be used to classify and enumerate the defects found.

I 4.7.1.2 Preparation for delivery. Each of the fully prepared shipping containers. including life preservers, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The list of defects as specified in table VI shall be used to enumerate the defects found.

4.7.2 Operation (carbon dioxide). The life preserver shall be placed in a flat condition with release pins removed, and pile fastener tapes separated. An inflation assembly conforming to MIL-I-23145, Type 11, Style III; a carbon dioxide cylinder conforming to MIL-C-25369, Type III; and a beaded handle assembly conforming to MIL-H-85478 shall be installed on each compartment for this test. The Individual compartments shall be Inflated by pulling the beaded handles to actuate the carbon dioxide cylinders and the preserver shall be observed for conformance to the requirements of 3.9.2.1. The preserver shall be Inspected for conformance to the requirements 3.9.1. Upon completion of this test and examination the buoyancy Inspection (4.7.5), if required, shall be performed while the preserver is Inflated with carbon dioxide. Upon completion of this test and inspection (or the buoyancy inspection) the preserver shall be completely deflated in the preparation for the pressure inspection (4.7.3).

4.7.3 Pressure. The carbon dioxide cylinders and inflator assembly shall be removed for this Inspection. The bladders shall be inflated with air (see 4.6.4) through their respective oral inflation assembles to a pressure of 5 PSIG. The air supply shall be securely shut off and after not less than 10 minutes the pressure shall be checked, and If necessary, readjusted to the original pressure of 5 PSIG. At the end of not less than 10 minutes after the readjustment period the respective bladders shall be inspected for conformance to the requirements of 3.9.2. Upon completion of this Inspection, the bladders shall be completely deflated and then Inspected for conformance to leakage requirements (3.9.3 and 4.7.4).

I 4.7.4 Leakage. The carbon dioxide cylinders and Inflator assemblies shall be removed for this inspection. Inflate both bladders with air (see 4.6.4) through their respective oral inflatlon assemblies to a pressure of 2 PSIG. The air shall be securely shut off and after not less than 15 minutes, the pressure checked and, If necessary, readjusted to the original pressure of 2.0 PSIG. The temperature and barometric pressue shall be recorded at this time. At the end of not less than four hours after the readjusted period, the pressure shall be measured and readjusted for any change in temperature or barometric pressure. The preserver shall meet the requirements of (3.9.3).

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4.7.5 Buoyancy. The buoyancy inspection, when required, shall be conducted in conjunction with the operation (carbon dioxide) inspection while the life preservers are still inflated with the carbon dioxide (see 4.7.2). The Inflated life preserver shall be attached to a 60-pound steel weight and placed in clean, fresh water having a temperature of 73 ± 5 degrees Fahrenheit (23 ± 3 degrees Centigrade). The preserver shall be inspected for conformance to the requirements of 3.9.4. Upon comletion of this inspection, the preserver shall be completely deflated and dried without the application of heat.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or Minimum Protection, as specified (see 6.2),

5.1.1 Level A. Each LPU-23B/P Life Preserver shall be packaged in accordance with MIL-P-116, Method lC-2(modified). Each life preserver cleaned in accordance with process C-1, shall be dusted with zinc sterate or talc, the minimum amount required to prevent the polychloroprene surfaces from adhering to each other in packaged condition. When talc is utilized, it shall be certified by the manufacturer as asbestosfree (see 6.2.2). Each flotation compartment shall be evacuated of air and each oral inflation valve, in closed condition, shall be tucked in place in the lobe. The life preserver shall be folded in accordance with drawing 68A73H1 and shall be completely wrapped In Grade A, 35-pound untreated Kraft paper conforming to UU-P-268. Every precaution shall be taken to prevent chafing by hardware or fittings. The wrapped preserver shall be packaged within a snug fitting fiberboard container conforming to PPP-B-636, Style CSSC, Type CF or SF, ldeather-Resistant Class, Variety SW, Grade W5c or W5s. The body joint and the top and bottom flaps shall be firmly glued together as specified in PPP-B-636. The fiberboard container shall not contain any metal fastening or stitches. All the seams and joints shall be sealed with water-resistant tape conforming to PPP-T-76, not less than two inches wide.

5.1.2 Minimum Protection. Each life preserver prepared and folded as specified in 5.1.1 shall be individually packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing. Packing shall be Level A, B, or Commercial, as specified (see 6.2). Shipping containers, Insofar as possible, shall be uniform in size and shape and of minimum cube and tare weight.

5.2.1 Level A. Six life preservers, packaged as specified in 5.1.1 or 5.1.2 shall be packaged as specified In 5.2.2, except that the fiberboard container shall be Heather-Resistant Class, Variety SH, Grade V3C and V3S. In addition, each container shall be reinforced with flat steel strapping or taped banding in accordance with the appendix to PPP-B-636.

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5.2.2 Level B. Six life preservers packaged as specified in 5.1.1 or 5.1.2 shall be packed edgewise into a close fitting fiberboard container conforming to PPP-B-636, Style CSSC, Vareity SW, Grade 275, Type-optional. The container body joint and top and bottom flaps, closure thereof shall be accomplished by glueing. The container shall be reinforced (strapped) by two non-metallic bands, applied in accordance with applicable procedures in he appendix to PPP-B-636. No metal staples or stitches are permitted on the container.

5.2.3 Minimum Protection. The packaged life preservers shall be packed with exterior type shipping containers in a manner that will insure safe transportation to the lowest rate to the point of delivery. The container shall be reinforced (strapped with two non-metallic bands. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking. In addition to any special markings required by the contract or order, the interior and exterior containers shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture (month and year).

6. NOTES

6.1 Intended use. The LPU-23B/P life preserver together with the automatic inflation device is Intended for use as emergency and life support equipment by personnel in ejection seat aircraft forced down at sea. This specification does not include the automatic Inflation device which must be ordered separately.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents shall specify the following:

a. Title, number and date of this specification, including any amendments.

b. Applicable drawings, including revisions.

c. Government part number, national stock number and quantity

d. Applicable Qualified Products List

e. Whether First Article inspection is waived (see 4.4.1)

f. Name and address of the First Article Inspection facility (see 4.4.1); and the name and address of the Government activity responsible for conducting the First Article inspection program (see 4.4.2)

9. Name and address of the Quality Conformance Verification Inspection facility (see 4.5.1.3), and the name and address of the Government activity responsible for conducting the Quality Conformance Verification inspection program (see 4.5.1.3)

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a. The edge of a flap

h. Selection of applicable levels of packaging and packing (see 5.1 and 5.2)

i. Whether any special markings are required (see 5.3)

j. Certificate of compliance for the age of the materials and components, excluding the metallic parts (see 3.3 and 6.2.2)

k. Selection of Qualified Conformance Verification Inspection samples to be performed by the cognizant Government Quality Assurance Representative/Specialist (see 4.5.1.3).

6.2.2 Contract data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), Incorporated Into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410.6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purcahse order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.	Options
(a) 3.3 and 6.2.1j	Certificate of Compliance for the Age of materials and Components.	DI-E-2121-Certificate of Compliance.	----- --
(b) 5.1.1	Certificate that TALC is Asbestos-free.	DI-E-2121-Certificate of Compliance.	----- --

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 Patterns. The pattern drawing for the life preserver casing shall be furnished by the contracting officer to the contractor for use in cutting working patterns (see 3.4.1.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for Inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the product that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington, DC 20360; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Development Center, Warminster, PA 18974, Attention: Code 6031. Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Naval Air Development Center (Code 6031) indicating a

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intended purpose

date on which the samples can be forwarded, and requesting an authorization number to accompany the samples, and the name and address of the Qualification Inspection Facility.

6.3.1 Qualification Inspection report. When requested, the manufacturer shall submit an inspection report in accordance with SD6, Provisions Governing Qualification.

6.4 First article. When a first article Inspection is required, the items should be a first production item. The first article should consist of three units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article Inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Subject term (keyword) listing.

Carbon dioxide
Bouyancy
Oral inflation

6.6 Changes from previous issue Asterisks (or vertical lines) are not used In this revision to Identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - AS
(Project No. 4220-N313)

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TABLE I. Qualification examinations and tests. (See 4.3.2)

Inspection	Paragraph	
	Requirement	Method
Operation (carbon dioxide)	3.9.1	4,7.2
Buoyancy	3.9.4	4.7.5
Pressure	3.9.2	4.7.3
Leakage	3.9.3	4.7.4
Visual examination	3.4 through 3.4.6, 3.7 through 3.8.2, 3.10, and applicable drawings	4.7.1
Dimensional check	3.6 and applicable drawings	4.7.1.1

TABLE II. First article examinations and tests. (See 4.4.3)

Inspection	Paragraph	
	Requirement	Method
Operation (carbon dioxide)	3.9.1	4.7.2
Buoyancy	3.9.4	4.7.5
Pressure	3.9.2	4.7.3
Leakage	3.9.3	4.7.4
Visual examination	3.4 through 3.4.6, 3.7 through 3.8.2, 3.10, and applicable drawings	4.7.1
Dimensional check	3.6 and applicable drawings	4.7.1.1

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TABLE III. Sample size, acceptance criteria, tests, and examinations of the components and the life preservers and preparation for delivery. (See 4.5.1.2)

Inspection	Paragraph		Sample Size	Acceptance Criteria 1/
	Requirement	Method		
Visual examination	Section 3 and applicable drawings	4.7.1	a. Every life preserver for critical defects	a. Reject all units with any critical defect.
			b. Inspection Level II 1/ for minor defects	b. An acceptable quality level of 15.0 defects per 100 units for minor defects
Operation 2/	3.9.1	4.7.2	Every life preserver	Reject all defective units
Pressure 3/	3.9.2	4.7.3	Every life preserver	Reject all defective units
Leakage 2/	3.9.3	4.7.4	Every life preserver	Reject all defective units
Buoyancy 2/	3.9.4	4.7.5	Inspection Level S-3 1/	An acceptable quality level of 1.0 defects per 100 units
Dimensional check	3.6 and applicable drawings	4.7.1.1	Inspection Level S-3 1/	An acceptable quality level if 10.0 defects per 100 units
Preparation for delivery	Section 5	4.7.1.2	Inspection Level S-2 1/	An acceptable quality level of 2.5 defects per 100 units

1/ The sample size shall be based upon the applicable sample size code letter corresponding to the specified level of MIL-STD-105.

2/ These inspections shall be performed in sequence. The operation (carbon dioxide) Inspection shall be performed first, then the buoyancy inspection (when required) and then the leakage inspection. The results of the inspection shall be identifiable by the assigned serial numbers which shall be marked on the life preserver as specified in 3.8.2.

3/ Pressure inspection may be performed at any time prior to final assembly of the life preserver.

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TABLE IV. Classification of defects for visual examination of the life preserver. (See 4.7.1.1)

Critical	Minor
1. Any non-specified hole, cut, patch or burn.	201. Stitching and sewed seam construction, not as specified. 3/
2. Any fabric damaged, bruised abraded, containing imperfections, or is otherwise defective. 1/	202. Any spot or stain. 3/
3. Any stitching in the Inflatable section of the life preserver.	203. Any cut edge of the uncoated nylon fibrous materials not seared or containing sharp edges.
4. Any grommet missting, mislocated or improperly clinched. 2/	204. Any snap fastener missing, mislocated, improperly clinched, or otherwise defective. 3/ 4/
5. Inflation assembly or manifold stem assembly not in accordance with requirements, not located as specified or Inoperable.	205. Oral inflation assembly or components not specified type; oral valve locked in the closed position.
6. Any seam separating in the inflatable section.	206. Oral inflation assembly or components not specified type; oral valve not locked in the closed position.
7. Any inflatable section seam construction which does not meet the minimum requirement specified.	207. Any metal component or component improperly finished or containing nicks, burrs, dents, sharp edges or rough surfaces.
8. Any seam tape not applied to seams as specified. 2/	208. Color of any component, not as specified.
9. Any channels, or voids in any seam on the inflatable section.	209. Any required marking missing, illegible, incomplete, incorrect or improperly located.
0. Inflatable sections and casing assembly not joined, as specified.	210. Any inspection record patch or warning tag missing, Illegible, incomplete, Incorrect or improperly located.
1. Oral inflation assembly mislocated, blend, distorted or Inoperable. Oral tube clamp not Installed securely. 2/	

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TABLE IV. Classification of defects for visual examination of the life preserver. (See 4.7.1.1) -Continued

Critical	Minor
12. Any component, component part or required operation omitted; or any operation improperly performed, not herein classified. 2/	211. Excessive wrinkling in any cementing operation. 3/
13 Any component not as specified or any defect of a component or assembly not herein classified. 2/	212. Any inflatable section seam construction which exceeds the maximum requirements specified. 3/
	213. Cement on the cloth surfaces around the patches, attachments, seams, seam tape not visible or in excess of requirement. 3/
	214. Any clot or mass of adhesive.
	215. Overlap of seam tape in excess of requirements.
	216. Any seam tape, patch, or attachment separating.

1/ Any defect shall be classified critical if it appears in MIL-C-19002 fabric. No more than two or a combination of two contiguous broken or missing picks, tight ends, knots or burrs, or creases are permitted in a "6 x 3" area of the finished casing of MIL-C-81814 fabric. Baggy, ridgy, wavy and defects which allow light transmission shall be scored as critical in the casing fabric (MIL-C-81814).

2/ The defect shall be classified Critical when it seriously affects serviceability or function; otherwise it is to be classified Minor.

3/ The defect shall be classified as Minor when it does not seriously affect serviceability or function; otherwise, it is to be classified Critical.

4/ The snap fasteners shall be checked for proper function and attachment.

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TABLE V. List of defects for finished dimensions. (See 4.7.1.1)

EXAMINE	DEFECT
Measure the life preserver, components, and the casing components.	Any measurement deviating from the dimensions and tolerances as specified in 3.4 through 3.8.3, and applicable drawings shall be enumerated as dimensional defect.

TABLE VI. List of defects for preparation for delivery. (See 4.7.1.2)

ITEM	DEFECT
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence, or method of application; markings not the same on the interior and exterior containers.
Packaging and Packing materials	Any non-conformance component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, container flaps, loose strappings, etc.; bulging or distortion of the containers; unit container contains metal fastening or stitches.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirements; item not folded as specified.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-85347A (AS)	LIFE PRESERVER, INFLATABLE, CEMENTED AIRCREWMAN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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